

Assignment 1: BGP Routing with ns-3

Course: Advanced Computer Networks(COL724/7524)

Overview

This assignment introduces the idea of inter-AS (Autonomous System) routing using a simplified *BGP* simulation in ns-3.

You are provided with a baseline script (`bgp.cc`) that sets up three routers (AS1–AS2–AS3) connected in a line and sends UDP traffic across them.

Your tasks are to extend this baseline into more realistic inter-domain situations by adding links, policies, and failures. Each part after **A** must be implemented as a **separate script** (e.g., `bgp_partB.cc`, `bgp_partC.cc`, `bgp_partD.cc`).

Part 0 (Background Research – Real BGP Data)

Before running any simulations, investigate **real BGP connectivity of IIT Delhi** using online BGP monitoring tools.

- (a) Visit any public BGP looking glass / route server (e.g., `bgp.he.net`, `RIPEstat`, or `NLNOG Looking Glass`).
- (b) Search for IIT Delhi's IP prefix (for example, `103.27.9.0/24`) or ASN.
- (c) Record:
 - Which **upstream ISPs** announce IIT Delhi's routes to the global Internet.
 - From which providers those ISPs themselves receive announcements (follow the chain at least three step further).
- (d) Summarize the findings in a short paragraph and include a screenshot of the BGP route view.

Deliverable: A 1–2 page writeup explaining IIT Delhi's real-world BGP peers and the upstream connectivity hierarchy you observed.

Part A (Baseline – Provided)

- Run the given `bgp.cc`.
- Confirm packets flow along the path **AS1 → AS2 → AS3**.
- Observe and explain the console log.
- Submit the log output along with the generated file `bgp-lite-results.xml`.

Note: This part is provided as a warm-up. You do not need to modify the code.

Part B (New Script Required)

Add a direct peering & compare paths

- Create a new script (`bgp-partB.cc`).
- Add a **direct link between AS1 and AS3**, parallel to the AS1–AS2–AS3 path.
- Configure the new link to have **higher bandwidth and lower delay**.
- Recompute routes and run the simulation.
- Show which path the traffic now takes and explain why.

Deliverables: script + console log excerpt + XML results.

Part C (New Script Required)

Simulate a policy preference (Local-Pref analogy)

- Create a new script (`bgp-partC.cc`).
- Even though the AS1–AS3 link is faster, implement a **policy** so AS1 prefers sending traffic via AS2.
- This can be done by:
 - Adding a **static route** to force traffic via AS2, or
 - Adjusting link costs/attributes so the AS1–AS2–AS3 path is chosen.
- Re-run and demonstrate that packets follow AS1 → AS2 → AS3.

Deliverables: script + short explanation of how you enforced the policy + logs.

Part D (New Script Required)

Failure and recovery study

- Create a new script (`bgp-partD.cc`).
- Introduce a **link failure** (choose AS1–AS3 or AS2–AS3).
- Failure should start at `t = 5s` and last for 2 seconds.
- Ensure that if an alternate path exists, traffic continues; if not, show the outage.
- Use FlowMonitor to measure:
 - Packets delivered vs lost
 - Delay before/during/after failure

Deliverables: script + short timeline/analysis of results.

Part E (New Script Required)

Dual-path rerouting study

- Create a new script (`bgp-partE.cc`).
- Design a 4-router topology with two distinct paths between AS1 and AS3:
 - Path A: R1–R2–R3
 - Path B: R1–R4–R3
- At time $t = 5s$, fail the link R2–R3.
- At time $t = 7s$, restore the failed link.
- Send continuous traffic from R1 to R3 throughout the experiment.
- Use FlowMonitor to measure:
 - Packet delivery and loss during failure and recovery
 - Route changes observed after link failure and restoration

Deliverables: new script + results (graphs or tables of delivery/loss and route changes).

Submission Guidelines

- Submit a single compressed archive (`.zip`) containing:
 - All source code files for Parts B, C, D, and E.
 - Output logs, XML results, and graphs/tables as required by each part.
 - Your write-up for Part 0.
- The zip file must be named in the following format: `COL<course_code>_assignment1_1_<entry_number>.zip` for first submission and `COL<course_code>_assignment1_2_<entry_number>.zip` for second submission (for example: `COL724_assignment1_1_2025MT1234.zip`).
- Ensure that each script is clearly named (`bgp-partB.cc`, `bgp-partC.cc`, etc.) and runs without modification in the provided ns-3 environment.

Administrative Details

Deadline: 03/09/2025 for Part 0, A, B, C

Deadline: 09/09/2025 for Part D, E

Work policy: This assignment must be completed *individually*. Collaboration on ideas is permitted at a high level, but sharing code or write-ups is strictly prohibited.